

Discrete-Phase Orthogonal Waveform Design for MIMO Radar via Soft Quantization Network

Hezhe Jia, Jun Pan, Xin Tai, Yongfeng Zuo, Guo Jing, Kai Zhong, Huiyong Li

Abstract—The design of discrete-phase orthogonal waveforms is crucial for multiple-input multiple-output (MIMO) radar. Existing methods generally fall into two categories: model relaxation methods, which are typically limited to optimizing **integrated sidelobe level (ISL)** with introducing relaxation errors, and search-based methods, which often suffer from high computational complexity. The optimization problem of orthogonal waveforms is modeled as a comprehensive optimization of **ISL** and **peak sidelobe level (PSL)** under discrete phase constraints. This optimization problem is non-smooth and non-convex, making it challenging to solve. We observe that nonlinear functions can approximate quantization functions, and deep learning can optimize non-convex, non-smooth problems. Given these findings, we propose the soft quantization network (SQN), combining residual networks and a soft quantization layer, to address this problem. The simulation results show that the waveforms designed by our proposed method exhibit better orthogonality than existing method and require less computation time when optimizing long sequences.

Index Terms—MIMO radar, orthogonal waveform design, discrete phase, soft quantization network (SQN).

I. INTRODUCTION

Designing orthogonal waveforms based on discrete phase encoding is a key issue in the research of multiple-input multiple-output (MIMO) radar [1]–[5]. Lower autocorrelation sidelobes help in detecting weak targets and reducing the probability of interception, while lower cross-correlation improves waveform diversity [6]–[11]. Phase-coded waveforms are favored in radar systems due to their constant modulus property, which prevents transmitter overload and nonlinear distortion [12]–[16]. Additionally, discrete-phase waveforms can simplify signal processing, reduce costs, and facilitate synchronization and decoding. As a result, discrete-phase coding waveform design has garnered widespread attention.

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Existing methods can be categorized into two types: (1) model relaxation-based methods, which introduce quantization errors; (2) search-based methods, which typically require extensive computational time.

In the first category, model relaxation based methods have been extensively applied. For the selection of the objective function, most methods focus on ISL optimization. Algorithms within the alternating optimization method have been developed in [17], [18]. Li *et al.* [17] focused on ISL and the designed waveform performed well in terms of average correlation without considering the peak level. Considering the peak level, Bu *et al.* [18] introduced auxiliary variables and Eamaz *et al.* [12] proposed relaxed power method-like iterations to design the discrete-phase unimodular sequences with low weighted ISL. Both methods obtained good autocorrelation, but neither considered cross-correlation. Considering both autocorrelation and cross-correlation, methods based on the majorization-minimization framework have been proposed in [19]–[21]. However, these methods focus on optimizing ISL while neglecting PSL. The methods proposed in [22], [23] are capable of optimizing PSL. Pishrow *et al.* [22] proposed an iterative algorithm using sequential rank-one constraint relaxation to optimize either ISL or PSL, but it is computationally expensive. Ehsan Raei *et al.* [23] proposed block successive upper bound minimization to design set of sequences with different properties, the proposed method can only optimize either PSL or ISL efficiently. However, these methods cannot jointly optimize PSL and ISL, and they introduce relaxation errors when quantizing phases.

In the second category, search-based optimization methods have been applied. The joint optimization of ISL and PSL has been studied in [24], [25]. In [24], Liu *et al.* proposed a multi-objective quantum genetic algorithm to jointly optimize PSL and ISL under discrete phase constraints, but the results showed that the ISL performance was unsatisfactory. In [25], Wang *et al.* proposed a novel waveform design algorithm based on intelligent ions motion optimization to optimize a weighted objective function incorporating both PSL and ISL. However, their approach only considered the average sidelobe level of the waveform, whereas the peak sidelobe level is often a more accurate indicator of orthogonality. As the number of coded bits increases indefinitely, discrete phase coding transitions into continuous phase coding. In this case, a non-gradient optimization method based on the **block coordinate descent (BCD)** [9], [10], [26]–[28] framework is used to optimize phase sequences for various objectives. The BCD method reduces the loss function by alternately optimizing

parameters, making it suitable for jointly optimizing PSL and ISL. However, for long sequences, BCD requires significant computational time.

Unlike existing studies, we notice that deep learning networks possess strong fitting capabilities, making them well-suited for solving non-convex and non-smooth optimization problems. Moreover, nonlinear functions can effectively model hard quantization, aligning naturally with our target. Based on this insight, we propose the soft quantization network (SQN) for discrete-phase orthogonal waveform design. We model the optimization problem as a joint optimization of PSL and ISL. First, a randomly initialized phase is processed through our designed residual network module, followed by a soft quantization layer to obtain an approximately discrete phase. The loss function is then computed and optimized using the Adam optimizer to update the parameters of both the residual network and the soft quantization layer. Finally, the approximate discrete phase undergoes hard quantization to produce the final discrete phase. Our main contributions are as follows:

- 1) We propose a learning-based soft quantization network for discrete phase coding, which can optimize the comprehensive performance metrics of orthogonal waveforms.
- 2) Our proposed method achieves an autocorrelation side-lobe level up to 6.64 dB lower and a cross-correlation level up to 2.22 dB lower than those of BCD.
- 3) Simulation results show that the proposed method requires less computational cost when optimizing long sequences.

II. SYSTEM MODEL AND PROBLEM FORMULATION

Consider a narrow band MIMO radar system equipped with M colocated transmit antennas, each antenna transmits N phase-encoded pulse signals, denoted as

$$\mathbf{s}_m = [s_m(1), \dots, s_m(n), \dots, s_m(N)]^T \in \mathbb{C}^N, \quad (1)$$

where $m = 1, \dots, M, n = 1, \dots, N$, the single pulse signal of $\mathbf{s}_m$ is constructed as $s_m(n) = e^{iy_m(n)}$, and $y_m(n)$ is the phase of that pulse signal. Construct the waveform matrix as $\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_n, \dots, \mathbf{s}_M] \in \mathbb{C}^{N \times M}$, the corresponding phase matrix is $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_m, \dots, \mathbf{y}_M] \in \mathbb{R}^{N \times M}$. The number of discrete phases is set as B , thus $y_m(n) \in 2\pi \times \{0, \dots, \frac{i}{B}, \dots, \frac{B-1}{B}\}$.

The orthogonality of waveforms is measured by correlation functions, and in this paper we study the aperiodic correlation functions of waveforms. Therefore, the cross-correlation function between waveform $\mathbf{s}_m$ and waveform $\mathbf{s}_l$ in lag k can be calculated by the following formula:

$$r_{ml}(k) = \sum_{n=1}^{N-k} s_m(n) s_l^*(n+k) = r_{lm}^*(-k), \quad (2)$$

where $m, l = 1, \dots, M, -N+1 \leq k \leq N-1$. When $m = l$, (2) is the autocorrelation function of the waveform $\mathbf{s}_m$.

PSL and ISL are measures of waveform orthogonality, which are given as

$$\text{PSL} = \max \left\{ \max_m \max_{k \neq 0} |r_{mm}(k)|, \max_{\substack{m,l \\ m \neq l}} \max_k |r_{ml}(k)| \right\}. \quad (3)$$

$$\text{ISL} = \sum_{m=1}^M \sum_{\substack{k=-N+1 \\ k \neq 0}}^{N-1} |r_{mm}(k)|^2 + \sum_{\substack{m,l=1 \\ m \neq l}}^M \sum_{k=-N+1}^{N-1} |r_{ml}(k)|^2. \quad (4)$$

According to the autocorrelation and cross-correlation, we refine (3) and (4) into APSL, CPSL, AISL, CISL, respectively, given by

$$\text{APSL} = \max_m \max_{k \neq 0} |r_{mm}(k)| \quad (5)$$

$$\text{CPSL} = \max_{\substack{m,l=1 \\ m \neq l}} \max_k |r_{ml}(k)| \quad (6)$$

$$\text{AISL} = \sum_{m=1}^M \sum_{\substack{k=-N+1 \\ k \neq 0}}^{N-1} |r_{mm}(k)|^2 \quad (7)$$

$$\text{CISL} = \sum_{\substack{m,l=1 \\ m \neq l}}^M \sum_{k=-N+1}^{N-1} |r_{ml}(k)|^2. \quad (8)$$

Low APSL signals enhance the detection of weak targets, while low CPSL signals improve multi-target detection resolution. Low AISL signals strengthen anti-jamming capabilities, and low CISL signals boost multi-signal coexistence performance. Therefore, considering the comprehensive optimization of the four metrics, the problem is modeled as

$$\begin{aligned} \min_{\mathbf{S}} \quad & L = \underbrace{\frac{(M-1)N}{2} l_1 \times \max(l_2 \text{APSL}, l_3 \text{CPSL})}_{\text{PSL}} \\ & + \underbrace{(M-1) \times l_4 \text{AISL} + l_5 \text{CISL}}_{\text{ISL}} \\ \text{s.t.} \quad & s_m(n) = e^{iy_m(n)}, \quad m = 1, \dots, M \text{ and } n = 1, \dots, N \\ & y_m(n) \in 2\pi \times \left\{ 0, \dots, \frac{i}{B}, \dots, \frac{B-1}{B} \right\}, \end{aligned} \quad (9)$$

The four metrics are harmonized to a consistent scale, and adjustable weighting coefficients $[l_1, l_2, l_3, l_4, l_5]$ are incorporated to allow flexible prioritization among the individual metrics.

III. THE PROPOSED METHOD

Since PSL optimization is a non-smooth process that cannot be addressed by gradient-based methods, most approaches focus solely on optimizing ISL. Additionally, the discrete phase constraint leads existing studies to either introduce relaxation errors or result in high computational complexity. In existing studies, BCD can address the comprehensive optimization problem with discrete phase constraints. However, it exhibits high computational complexity when optimizing long sequences.

In contrast to existing studies, the proposed method soft quantization network (SQN) designs a residual network module to fit the phase sequence and introduces a learnable soft quantization layer to approximate hard quantization, which is well-suited for this comprehensive optimization problem. The framework of SQN is shown in Fig. 1, followed by a detailed description of each part.

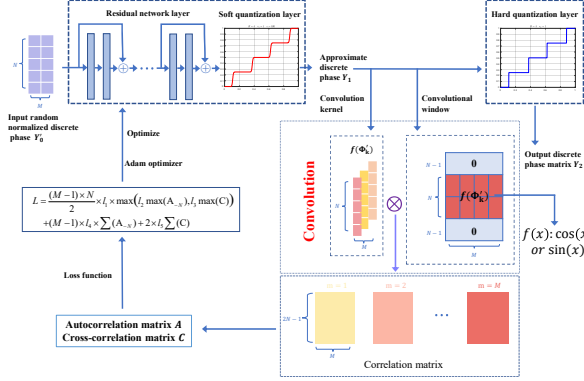


Fig. 1: The framework of SQN.

A. Residual Network Module

The input to this network is set as a randomly normalized discrete phase sequence $\mathbf{Y}'_0 = \text{vec}([\mathbf{y}_1, \dots, \mathbf{y}_m, \dots, \mathbf{y}_M]) \in \mathbb{R}^{NM \times 1}$, where the elements of the sequence $y'_m(n) \in \{0, \dots, \frac{i}{B}, \dots, \frac{B-1}{B}\}$.

Since the network input is a one-dimensional sequence, fully connected layers are used. To address issues such as gradient vanishing and explosion caused by an excessive number of fully connected layers, residual connections are introduced. Therefore, the mathematical representation of the residual block is as follows:

$$\begin{aligned} \hat{\mathbf{x}}_{2i-2} &= \begin{cases} \mathbf{x}_{2i-2} & h_1 = h_3 \\ \mathbf{x}_{2i-2} \times \hat{\mathbf{W}} + \hat{\mathbf{b}} & h_1 \neq h_3 \end{cases}, \\ \mathbf{x}_{2i-1} &= \tanh(\mathbf{x}_{2i-2} \times \mathbf{W}_{2i-1} + \mathbf{b}_{2i-1}), \\ \mathbf{x}_{2i} &= \tanh(\mathbf{x}_{2i-1} \times \mathbf{W}_{2i} + \mathbf{b}_{2i} + \hat{\mathbf{x}}_{2i-2}). \end{aligned} \quad (10)$$

where $\mathbf{x}_{2i-2} \in \mathbb{R}^{1 \times h_1}$ represents the input of the i residual block ($2i-1$ layer network) and the $i-1$ residual block ($2i-2$ layer network), $\mathbf{x}_{2i-1} \in \mathbb{R}^{1 \times h_2}$ is the output of the $2i-1$ layer network. $\mathbf{W}_{2i-1} \in \mathbb{R}^{h_1 \times h_2}$ and $\mathbf{b}_{2i-1} \in \mathbb{R}^{1 \times h_2}$ are respectively the weights and biases involved in training in the $2i-1$ fully connected layer. h_1 is the dimension of the input to this residual block, and h_2 is the number of neurons in the $2i-1$ fully connected layer. $\hat{\mathbf{x}}$ is the identity mapping, h_3 is the number of neurons in the $2i$ fully connected layer.

The input random phase sequence passes through the residual module, then is normalized to the range $[0, 1]$ using a sigmoid function before being fed into the soft quantization layer.

B. Soft Quantization Layer

Traditional quantization uses fixed thresholds for hard quantization, which can introduce significant errors during the process. In the soft quantization layer, we incorporate learnable thresholds and combine them with the objective function to achieve better quantization results. We notice that nonlinear functions can approximate quantization functions. Therefore we design a soft quantization layer which is a differentiable approximation method, it can simulate the effect of hard quantization while allowing gradients to be backpropagated through the layer. Its mathematical model is

$$Q_A(x) = \sum_{i=0}^{B \times r_n - 1} a_i [\tanh(c(x - \rho_i)) + 1] \quad (11)$$

where $B = 2^b$, $a_i = a = \frac{1}{2^{b+1}}$, b is the number of bits encoded in discrete phase, B is the number of phases, c denotes proximity to hard quantization. ρ_i is the threshold of the soft quantization function and is a learnable parameter. r_n reduces the gap between the quantization thresholds in the soft quantization layer, allowing it to more closely approximate hard quantization. The output of the soft quantization layer is

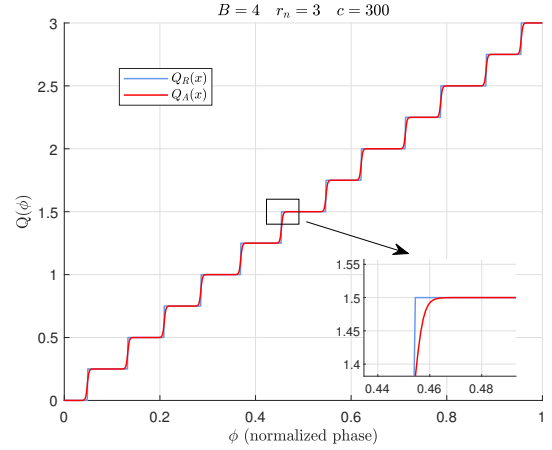


Fig. 2: $Q_A(x)$ and $Q_R(x)$.

an approximate discrete phase sequence, which, after passing through $Q_R(x)$, results in the discrete phase sequence. $Q_A(x)$ and $Q_R(x)$ are shown in Fig. 2.

$$Q_R(x) = \frac{1}{2^b} \sum_{i=0}^{r_n \times 2^b - 1} H(x - \rho_i) \quad (12)$$

where $H(x)$ is Heaviside function which is also called step function.

C. Loss Function Calculation through Convolution

The soft quantization function does not obstruct the back-propagation of gradients, allowing us to compute the correlation function of the approximate discrete phase sequence output by the soft quantization layer as the loss function. Since

$s_m(n) = e^{iy_m(n)} = \cos(y_m(n)) + i \sin(y_m(n))$, we rewrite (2) as

$$r_{ml}(k) = \sum_{n=1}^{N-k} s_m(n) s_l^*(n+k) = \sum_{n=1}^{N-k} e^{iy_m(n)} \times e^{-iy_l(n+k)} \\ = \sum_{n=1}^{N-k} \begin{pmatrix} \cos(y_m(n)) \times \cos(y_l(n+k)) \\ + \sin(y_m(n)) \times \sin(y_l(n+k)) \\ + i \left(\sin(y_m(n)) \times \cos(y_l(n+k)) \right. \\ \left. - \cos(y_m(n)) \times \sin(y_l(n+k)) \right) \end{pmatrix} \quad (13)$$

Noticing that the calculation method of each part is the same as convolution, so the correlation calculation can be performed as follows

$$\mathbf{r}_{ml} = \hat{\mathbf{p}}_l \otimes \mathbf{p}_m + \hat{\mathbf{q}}_l \otimes \mathbf{q}_m + i(\hat{\mathbf{p}}_l \otimes \mathbf{q}_m - \hat{\mathbf{q}}_l \otimes \mathbf{p}_m) \quad (14)$$

where $\otimes$ denotes convolution operations, $\mathbf{p}_{l(m)} = \cos(\mathbf{y}_{l(m)}) \in \mathbb{R}^{N \times 1}$, $\mathbf{q}_{l(m)} = \sin(\mathbf{y}_{l(m)}) \in \mathbb{R}^{N \times 1}$, $l, m = 1, \dots, M$, $\hat{\mathbf{p}}_{l(m)} = [0_{1 \times N-1}, \mathbf{p}_{l(m)}^T, 0_{1 \times N-1}]^T \in \mathbb{R}^{(3N-2) \times 1}$, $\hat{\mathbf{q}}_{l(m)} = [0_{1 \times N-1}, \mathbf{q}_{l(m)}^T, 0_{1 \times N-1}]^T \in \mathbb{R}^{(3N-2) \times 1}$. Therefore, $\mathbf{r}_{ml}$ can be computed through convolution.

On the basis of correlation calculation, we can get autocorrelation matrix and cross-correlation matrix

$$\mathbf{A} = [|\mathbf{r}_{11}|^2, \dots, |\mathbf{r}_{mm}|^2, \dots, |\mathbf{r}_{MM}|^2] \quad (15)$$

$$\mathbf{C} = [|\mathbf{r}_{12}|^2, \dots, |\mathbf{r}_{1m}|^2, |\mathbf{r}_{23}|^2, \dots, |\mathbf{r}_{(M-1)M}|^2], \quad (16)$$

where $\mathbf{A} \in \mathbb{C}^{(2N-1) \times M}$, $\mathbf{C} \in \mathbb{C}^{(2N-1) \times \frac{(M-1)M}{2}}$, $|\mathbf{r}_{ml}|^2 = (\hat{\mathbf{p}}_l \otimes \mathbf{p}_m + \hat{\mathbf{q}}_l \otimes \mathbf{q}_m)^2 + (\hat{\mathbf{p}}_l \otimes \mathbf{q}_m - \hat{\mathbf{q}}_l \otimes \mathbf{p}_m)^2$.

Thus APSL, AISL, CPSL, CISL can be calculated as

$$\begin{aligned} \text{APSL} &= \max(\mathbf{A}_{-N}) \\ \text{AISL} &= \sum(\mathbf{A}_{-N}) \\ \text{CPSL} &= \max(\mathbf{C}) \\ \text{CISL} &= 2 \times \sum(\mathbf{C}) \end{aligned} \quad (17)$$

where $\mathbf{A}_{-N}$ denotes the removal of the N th row of $\mathbf{A}$, i.e., eliminating the case where the lag $k = 0$. The loss function (9) can be further written as

$$\begin{aligned} L &= \frac{(M-1) \times N}{2} \times l_1 \times \max(l_2 \text{APSL}, l_3 \text{CPSL}) \\ &+ (M-1) \times l_4 \times \text{AISL} + l_5 \text{CISL} \\ &= \frac{(M-1) \times N}{2} \times l_1 \times \max(l_2 \max(\mathbf{A}_{-N}), l_3 \max(\mathbf{C})) \\ &+ (M-1) \times l_4 \times \sum(\mathbf{A}_{-N}) + 2 \times l_5 \sum(\mathbf{C}). \end{aligned}$$

D. Convergence Judgment

In the judgment of convergence, since the selection of discrete phases is limited, both the soft quantization loss function and the hard quantization loss function are considered. Define the maximum number of iterations D , the length of iteration intervals as E , the convergence factor θ , and record the

minimum soft quantization loss value l_{min}^i and the minimum hard quantization loss c_{min}^i within i intervals, where $\lfloor \cdot \rfloor$ denotes the downward rounding operation. The training ends when one of the following two conditions is satisfied.

- i. Notation $r_{min}^i = 0.5 \times (l_{min}^i + c_{min}^i)$, when $|(r_{min}^{j+1} - r_{min}^j)/r_{min}^j| < \theta$ ($1 \leq j < j+1 \leq \lfloor D/E \rfloor$) is satisfied, convergence is considered to be achieved, iteration is withdrawn, and the discrete phase sequence corresponding to the minimum hard quantized loss value is output.
- ii. D iterations end and the discrete phase sequence corresponding to the smallest loss value is output.

IV. NUMERICAL RESULTS

In this section, the orthogonality of the generated waveforms and the optimization time are evaluated. The metrics evaluated are maximum autocorrelation sidelobe (MAS), maximum cross-correlation sidelobe (MCS) and optimization time. To validate the effectiveness of the proposed method, BCD is used as a comparison method. MAS and MCS are defined as follows:

$$\text{MAS} = 20 \log_{10} \left| \frac{\max_m r_{mm}(k)}{N} \right|, k = -N+1, \dots, N-1 \quad (19)$$

$$\text{MCS} = 20 \log_{10} \left| \frac{\max_{m,l,m \neq l} r_{ml}(k)}{N} \right|, k = -N+1, \dots, N-1. \quad (20)$$

In the simulation analysis, the weights of the comprehensive objective function are set to $l_1 = l_2 = l_3 = l_5 = 1, l_4 = 3$, the residual network module consists of 5 residual blocks, the output dimension of each fully connected layer is set to $K = 128$, the value of r_n in the soft quantization layer is set to 10, the learning rate of Adam's optimizer δ is set to 0.00001, the maximum number of iterations is $D = 20000$, the length of the interval is $E = 1000$ and the convergence factor is $\theta = 0.001$.

Regarding the specification of waveform parameters, the number of waveforms is set to $M = 3$, with sequence lengths of $L = \{256, 512\}$, respectively. The number of phases is set to $B = \{4, 8, 16, 32\}$. The BCD method is configured with the same parameters.

Fig. 3 illustrates the convergence curve and the generated discrete phases for $M = 3$, $N = 256$, and $B = 4$. Fig. 3(a) shows the iterative loss behavior under the given conditions. "Soft" represents the loss of the approximate discrete phase, "hard" represents the discrete phase loss, and "min_hard" is the minimum loss recorded during training. A disparity exists between losses before and after hard quantization. Over time, the loss values converge, with "min_hard" stabilizing at a constant value. Fig. 3(b) shows the discrete phase encoding results under the given conditions.

Fig. 4 compares the MAS and MCS results of the SQN and BCD methods under various conditions. It can be observed that the waveforms designed by our proposed method exhibit lower correlation sidelobes than BCD. To quantitatively demonstrate

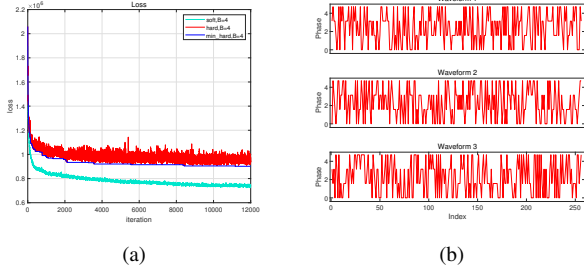


Fig. 3: Result for $M=3$, $N=256$, $B=4$ (a) Convergence curve (b) Generated discrete phases.

the superiority of our proposed method, the MAC, MCS, and optimization time are recorded in Table I.

For MAS, the sidelobe of SQN is up to 6.64 dB lower than that of BCD, with a minimum reduction of 0.60 dB. For MCS, SQN achieves a maximum reduction of 2.22 dB and a minimum of 0.54 dB compared to BCD. The simulation results demonstrate that the waveforms designed by our proposed method exhibit better orthogonality while also showcasing their adaptability under different parameter conditions. In terms of optimization time, SQN outperforms BCD for long-sequence waveform design. While BCD is faster for shorter sequences, its optimization time increases linearly with sequence length, making SQN a more efficient choice for longer sequences.

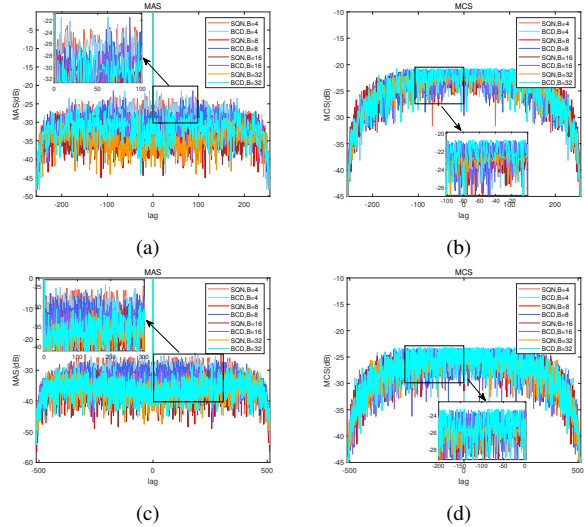


Fig. 4: Maximum autocorrelation and cross-correlation with $M = 3$, $N = [256, 512]$, $B = [4, 8, 16, 32]$ (a)MAS($N = 256$); (b)MCS($N = 256$); (c)MAS($N = 512$); (d)MCS($N = 512$)

V. CONCLUSION

In this work, the design of discrete-phase orthogonal waveforms for MIMO radar is studied. Unlike existing methods, we propose the SQN to optimize waveform phase sequences.

TABLE I
Maximum MAS sidelobe and MCS with $M=3$

N	B	MAS(dB)		MCS(dB)		Time(s)	
		SQN	BCD	SQN	BCD	SQN	BCD
256	4	-22.89	-21.46	-21.56	-20.80	48	27
256	8	-25.29	-23.59	-21.37	-20.83	68	56
256	16	-28.09	-21.45	-21.96	-20.92	106	112
256	32	-28.09	-25.15	-22.20	-20.75	198	227
512	4	-24.89	-24.29	-24.25	-23.36	84	79
512	8	-29.48	-26.54	-24.18	-23.48	158	161
512	16	-30.54	-27.50	-24.59	-23.35	247	325
512	32	-29.15	-28.46	-25.41	-23.19	465	652

Simulation results demonstrate that our method achieves better orthogonality and is well-suited for optimizing long sequences. In the future, we will focus on generative waveform design.

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